# Mostafa Alrahal

## Applied AI Engineer | LLM, RAG & Agentic AI Systems

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**Pittsburgh, PA** | **LinkedIn** | **Portfolio** | **GitHub**

Applied AI Engineer specializing in RAG pipelines, multi-agent systems, and multi-tenant AI applications using Python, LangGraph, and FastAPI. Contributed to Deloitte Middle East's AI Tax Assistant, developing retrieval workflows across thousands of proprietary tax and legal resources. Built reusable AI agent platforms that accelerated prototype delivery from days or weeks to hours, and developed a conversational financial assistant that reduced routine customer support tickets by 60%.

## Work Experience

**Maly Tech Ltd | Gen AI Engineer | Dubai, UAE** **Aug 2024 – May 2026**

- Contributed to the development and deployment of **Deloitte's AI Tax Assistant**, implementing RAG, LangGraph, and prompt engineering to ingest, index, and retrieve information from thousands of proprietary tax and legal resources for MENA tax guidance.
- Built **AI4ALL, a multi-tenant AI platform** for rapidly deploying and demonstrating customer-specific AI agents and prototypes, using Python, Google ADK, agent orchestration, tool calling, and prompt engineering; reduced the time required to deliver new AI prototypes from days or weeks to hours.
- Developed **Luna, a conversational AI financial assistant,** leveraging RAG, prompt engineering (Few-Shot, ReAct), and context-injection to deliver safe, personalized answers to financial inquiries, reducing routine customer support tickets by 60%.

**Freelance | Software Developer** **Oct 2023 – Aug 2024**

- Developed and maintained production web applications using **Angular, Laravel, and REST APIs**, building reusable UI components and integrating backend services to support feature delivery.

## Projects

**Multi-Agent Travel Planner** **2026**

- Built a LangGraph **multi-agent travel planner** that orchestrates specialized agents and dynamically provisions destination-specific RAG knowledge bases when information is unavailable from the indexed knowledge base.

**Document Processing Pipeline** **2026**

- Designed and implemented a **multi-tenant data ingestion service** for use with RAG system, automating the parsing and vectorization of PDFs, DOCX, PPTX, TXT, and MD files using Docling, LlamaParse, and Jina's late chunking, which accelerated the creation of new agentic RAG applications.

**Graduation Project | Fruitful Tree Count Management** **2024**

- Fine-tuned custom **deep learning model** and developed **OpenCV computer-vision algorithms** to automate fruit-tree counting and land-cover detection from satellite imagery, achieving >90% accuracy.

**Academic AI Projects** **2021 – 2024**

- Built applied NLP, computer-vision, deep-learning, and machine-learning projects, including hand-gesture recognition, text classification, and income prediction using OpenCV and Scikit-learn, PyTorch.

## Education

**Damascus University | BS in Information Technology Engineering** **2025**

- **Specialization:** Artificial Intelligence & Natural Language Processing.
- **GPA: 3.78/4.0** (WES Verified U.S. Bachelor's Equivalent).

## Skills

- **AI & ML:** Generative AI, AI Automation, Multi-Agent Orchestration, Deep Learning (DL), Machine Learning (ML), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Natural Language Processing (NLP), Prompt Engineering, Text Embeddings, Tool Calling, Computer Vision (CV), Knowledge Graphs.
- **Frameworks & Libraries:** LangGraph, LangChain, Google ADK, PyTorch, Scikit-learn, OpenCV, Docling.
- **Tools:** Python, Java, Typescript, FastAPI, Laravel, Angular, PostgreSQL, MongoDB, Vector DBs (pgvector, Atlas), Graph DBs (Neo4j), Redis, GCP, Docker, Git, RESTful APIs, Streaming APIs (SSE, WebSockets).